System Requirements & Test Specification (SRS / STS): Supervisory Powertrain Safety Controller

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Target Domain: High-Voltage EV / Formula Hybrid Powertrain Safety Architecture

1. Scope & System Overview:

This specification defines the functional, diagnostic, and supervisory safety requirements for an embedded telemetry acquisition pipeline coupled with a Model-in-the-Loop (MIL) supervisory torque cutoff controller. The system monitors driver pedal demand (APPS) and high-voltage accumulator module temperatures to detect sensor faults and enforce fail-safe torque suppression during thermal excursion events.

2. Functional Requirements (FR):

FR-01 (Deterministic Sampling): The embedded acquisition node shall sample dual analog channels (APPS and pack temperature) at a deterministic update rate of 20 Hz ($50\text{ ms} \pm 2\text{ ms}$ jitter window).

FR-02 (Signal Conditioning): Analog sensor inputs shall be conditioned via a 5-tap moving average digital filter to attenuate high-frequency SAR ADC conversion noise prior to serial transmission.

FR-03 (Telemetry Parsing & Alignment): Downstream processing scripts shall parse serial telemetry frames, compute sampling jitter, and normalize hardware timestamps to a relative continuous time vector ($t_0 = 0.0\text{ s}$).

3. Safety & Diagnostic Requirements (SR):

SR-01 (APPS Plausibility Check): The controller shall continuously monitor the APPS analog rail. If the raw voltage drops below 0.15 V (ground short/open harness) or exceeds 3.15 V (VCC short), the firmware shall flag an out-of-range condition (ERR_APPS_OOR).

SR-02 (Thermal Over-Temperature Trip): When the monitored accumulator temperature equals or exceeds the calibrated safety limit ($T \ge T_{\text{trip}}$), the supervisory controller shall force the powertrain torque demand from active driver demand directly to 0.0%.

SR-03 (Dynamic Reaction Time): The supervisory shutdown command shall be asserted within one discrete sample step ($\Delta t \le 50\text{ ms}$) following thermal limit detection.

4. Verification & Traceability Matrix (VTM):

Req ID	Requirement Summary	Verification Method	Pass/Fail Criteria	Result
FR-01	20 Hz acquisition timing	Python pipeline timestamp delta evaluation	evaluationMean delta = 50.0 ms ± 2.0 ms	PASS
FR-02	5-Tap moving average filter	Moving average buffer code inspection & telemetry log	logSignal noise ripple attenuated	PASS
SR-01	APPS plausibility bounds	boundsRail excursion testing (<0.15vV, >3.15 V)	Sensor fault flag asserted	PASS
SR-02	Zero-torque thermal trip	Simulink Model-inthe-Loop Scope time trace	Torque demand drops to 0.0% at trip	PASS
SR-03	≤ 50 ms trip latency	latencyScope discrete time step delta analysis	Cutoff occurs on the immediate next step	PASS